

Ruey-An Shiu

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Education

Carnegie Mellon University <i>PhD in Electrical and Computer Engineering</i>	Pennsylvania, USA Sep. 2026 – (Expected) Jun. 2031
National Taiwan University <i>B.S. in Mathematics, B.A. in Economics (Honor), GPA: 4.10/4.30</i>	Taipei, Taiwan Sep. 2021 – Jun. 2026
University of California San Diego <i>University of California Education Abroad Program, GPA: 4.00/4.00 (Provost Honors)</i>	California, USA Fall 2024
2025 SLMath Summer Graduate School <i>Statistical Optimal Transport</i>	California, USA Jun. 9 th – Jun. 20 th , 2025

Publication

Ruey-An Shiu and Hsuan-Wei Lee. “Shaping network structures in evolutionary games for stronger cooperation.” *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 481(2322), 20250346 (2025).

Ruey-An Shiu and Chun-Hsiung Hsia. “On the Synchronization of ReLU Version of Dense Networks.” *Chaos, Solitons and Fractals: the Interdisciplinary Journal of Nonlinear Science, and Nonequilibrium and Complex Phenomena*, Volume 202, Part 1, 117444 (2026).

Tony Feng, Trieu Trinh, Garrett Bingham, Jiwon Kang, Shengtong Zhang, Sang-hyun Kim, Kevin Barreto, Carl Schildkraut, Junehyuk Jung, Jaehyeon Seo, Carlo Pagano, Yuri Chervonyi, Dawsen Hwang, Kaiying Hou, Sergei Gukov, Cheng-Chiang Tsai, Hyunwoo Choi, Youngbeom Jin, Wei-Yuan Li, Hao-An Wu, **Ruey-An Shiu**, Yu-Sheng Shih, Quoc V. Le, Thang Luong. “Semi-Autonomous Mathematics Discovery with Gemini: A Case Study on the Erdős Problems.” *arXiv preprint arXiv:2601.22401*. (2026)

Ruey-An Shiu and Parinaz Naghizadeh. “Coordination games on multiplex networks: Consensus, convergence, and stability of opinion dynamics.” *arXiv preprint arXiv:2603.07633*, (2026).

Conference

Ruey-An Shiu and Parinaz Naghizadeh. “Opinion Dynamics in Multi-Layer Social Networks: Consensus and Convergence in Coordination Games.” *The 61st Annual Allerton Conference on Communication, Control and Computing (Allerton’25)*, (2025).

Daniel Wei-Chung Miao, Xenos Chang-Shuo Lin, **Ruey-An Shiu**. “Exact Analytical Option Pricing Formula Under a Jump-Diffusion Model with Hawkes Intensity.” *Association of Asia-Pacific Operational Research Societies (APORS) Youth Forum*, (2025).

Research Experience

2026 Globalink Research Internship, University of Waterloo ○ Advisor: Prof. Roberto Guglielmi ○ Topic: Mathematical Modelling and Control of Wildfire	Summer 2026
Research Assistant, Department of Economics, NTU ○ Advisor: Prof. Hung-Jen Wang ○ Topic: Distributional Uncertainty in Stochastic Frontier Models	Sep. 2025 – Jun. 2026
Research Assistant, College of Health, Lehigh University (Remote) ○ Advisor: Prof. Hsuan-Wei "Wayne" Lee ○ Topic: Evolutionary Game Theory and Mathematical Epidemiology	Aug. 2024 – Present
Research Assistant, Financial Engineering Research Lab, NTUST	Aug. 2023 – Jan. 2026

- *Advisor:* [Prof. Daniel Wei-Chung Miao](#)
- **Topic:** Variants of the Jump-Diffusion Model for Option Pricing
- 2025 Student Summer Research Fellowship Program, University of Chicago** Summer 2025
- *Advisors:* [Prof. Haifeng Xu](#), [Prof. Jibang Wu](#)
- **Topic:** Robust Principal-Agent Problems: A General Framework
- Independent Study, UCSD (Remote)** Jan. 2025 – Sep. 2025
- *Advisor:* [Prof. Morris Ang](#)
- **Topic:** Random Conformal Geometry
- Member of the Multi-agent Intelligence and Decision Systems Lab, UCSD** Fall 2024
- *Advisor:* [Prof. Parinaz Naghizadeh](#)
- **Topic:** Coordination Games on Multiplex Networks
- 2024 Undergraduate Research Project Scholarship, NSTC** Jul. 2024 – Feb. 2025
- *Advisor:* [Prof. Daniel Wei-Chung Miao](#)
- **Topic:** Time-Heterogeneous Double-Exponential Jump-Diffusion Model for Option Pricing
- 2023 NCTS Undergraduate Research Program, NCTS** Oct. 2023 – Jun. 2024
- *Advisor:* [Prof. Chun-Hsiung Hsia](#)
- **Topic:** Collective Phenomena in Dynamical Systems Governed by Differential Equations
- 2023 Undergraduate Summer Research Program, NCTS** Summer 2023
- *Advisors:* [Prof. Gi-Ren Liu](#), [Prof. Yuan-Chung Sheu](#)
- **Topic:** Central Limit Theorems and Stein's Method for Gaussian Approximations

Awards

2026: Carnegie Institute of Technology Dean's Fellowship
2025: International Federation of Operational Research Societies (IFORS) Travelling Fellowship
2025: NSTC College Student Research Creativity Award
2025: National Taiwan University Bachelor's Thesis Award
2025: Shih Kung-Sing Outstanding Thesis Award
2025: National Taiwan University Overseas Internship Scholarship
2024: Taiwan Mathematical Society Outstanding Students Prize
2024: Chow Hung-Ching Scholarship
Fall 2023, Spring 2024, Spring 2025: Academic Achievement Award (Top 5%)
2023: Finance and Mathematics Joint Scholarship
2023: National Taiwan University Voyage of Aspirations Scholarship for Exchange Students
2019: Asian Pacific Mathematics Olympiad (APMO) Bronze Award

Teaching Experience

Fall 2023, Fall 2024, Fall 2025 : Teaching Assistant, Stochastic Processes (FIN7050)
Fall 2023: Teaching Assistant, Introduction to Ordinary Differential Equations (MATH2217)
Spring 2024: Teaching Assistant, Introduction to Partial Differential Equations (MATH2218)

Computational Skills / Other

Programming: R, Python, C++, MATLAB, Julia
Languages: Mandarin (native), English (fluent)